

HOW IMPORTANT ARE CAPITAL AND TOTAL FACTOR PRODUCTIVITY FOR ECONOMIC GROWTH?

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We examine the relative importance of the growth of physical and human capital and the growth of total factor productivity (TFP) using newly organized data on 145 countries that spans more than 100 years for 23 of these countries. For all countries, only 14% of average output growth per worker is associated with TFP growth. We use priors from theories to construct estimates of the relative importance of the variances of aggregate input growth and TFP growth across countries. Much of the importance of the variance of TFP growth across countries is associated with negative TFP growth. (JEL O47, O50, O57, O30, N10)

How much of growth in output per worker is associated with growth in physical and human capital per worker, and how much is due to technology, institutional change, and other factors? An economy's output is a positive function of physical and human capital given the technology. Assumptions of constant returns to scale and competitive factor markets make it possible to calculate the growth rate of output implied by the growth of physical

and human capital; deviations of actual output from this implied growth rate are due to changes in technology, institutional change, failure of the twin assumptions of constant returns to scale and competitive factor markets, and other factors. These deviations are called growth in total factor productivity (TFP), although these deviations include much more than what is suggested by the word *productivity* and probably are more fairly called the "residual" or "Solow residual" in growth.

This type of analysis, called growth accounting, preceded the theoretical contributions to growth theory by Solow (1956) and Swan (1956), but many more publications succeeded them. Abramovitz (1956) found that only 10% of output growth per person in the United States from 1869–78 to 1944–53 is associated with growth of factors of production, and 90% of output growth is associated with growth of TFP. Solow (1957) found that the accumulation of physical capital accounts for roughly 12% of output growth per hour worked in the United States from 1900 to 1949 with the remaining 88% attributed to growth of TFP. Although later work has reduced this unexplained residual, it is

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ABBREVIATIONS

GDP: Gross Domestic Product
NIC: Newly Industrialized Country
PPP: Purchasing Power Parity
R&D: Research and Development
TFP: Total Factor Productivity

far from zero in work by Kendrick (1961), Denison (1985), Jorgenson et al. (1987), Maddison (1995), Klenow and Rodriguez-Clare (1997a), Jones (1997), and Abramovitz and David (2000). In short, such estimates indicate that the part of economic growth associated with growth of physical and human capital is dwarfed by the unexplained part.

The purpose of this article is to estimate the relative importance of physical and human capital growth and TFP growth for output growth using a more comprehensive data set than has been previously available. Our data set covers more countries for a longer period than other data sets. It includes the growth of human capital, as do some data sets, but they are less comprehensive than ours. Our computations are similar to those presented by Abramovitz (1956), Solow (1957), Kendrick (1961), Denison (1985), and others.

We find that growth in TFP contributes modestly to the average performance of output growth across all countries. We find that weighted-average TFP growth is only about 0.22% per year, which is about 14% of growth of output per worker. Fourteen percent is far from previous estimates of 50% or more of growth of output per worker. A simple unweighted average of TFP growth across these countries actually is negative: -0.81% per year. This means that if one of our 145 countries is chosen at random with equal probability, the expected growth rate of TFP is -0.81% per year. This hardly is suggestive of technological change, unless one thought that much of recent history is technological regress. This is improbable, and we think that this decline in TFP is most likely due to institutional retrogression and disruptive events such as armed conflicts.

This startling overall finding, however, masks a far more interesting tale by countries, which are aggregated into regions for some of our analysis. TFP growth is 34% of output growth per worker for the Western countries including the United States; 26% for Southern Europe; and 26% for newly industrialized countries (NICs). Although not on the order of 50% or more, this is not essentially zero either. On the other hand, Sub-Saharan Africa and the Middle East have negative TFP growth. Something more than introduction of new technology is necessary to explain much of these data.

Even though TFP's contribution to average output growth is modest, the variance of output growth per worker across all countries is more closely associated with the variance of TFP growth than with the variance of physical and human capital growth. As for the average growth rates, the analysis by regions reveals patterns. This predominance of the importance of variation of TFP growth is confined to Central and Eastern Europe, Sub-Saharan Africa, and Latin America. For the Western countries, Southern Europe, and the NICs, variation of growth of physical and human capital per worker may account for 90% or more of the variation of output growth per worker. The predominance of TFP growth for explaining the variance of output growth for some regions may seem contradictory to the low mean of TFP growth, but it is not. The low mean of TFP growth and nontrivial variance of TFP growth suggest that negative TFP growth is an important part of the variation in growth across countries, a conclusion reinforced by the analysis by regions.

The structure of the article is simple. The first section briefly discusses the logic of the calculations and the analytical underpinnings of our new data set.¹ The second section presents the analysis of mean growth rates. The third section presents our analysis of the variance decomposition across countries including the sensitivity of the analysis to the choice of time periods. The article ends with a brief conclusion.

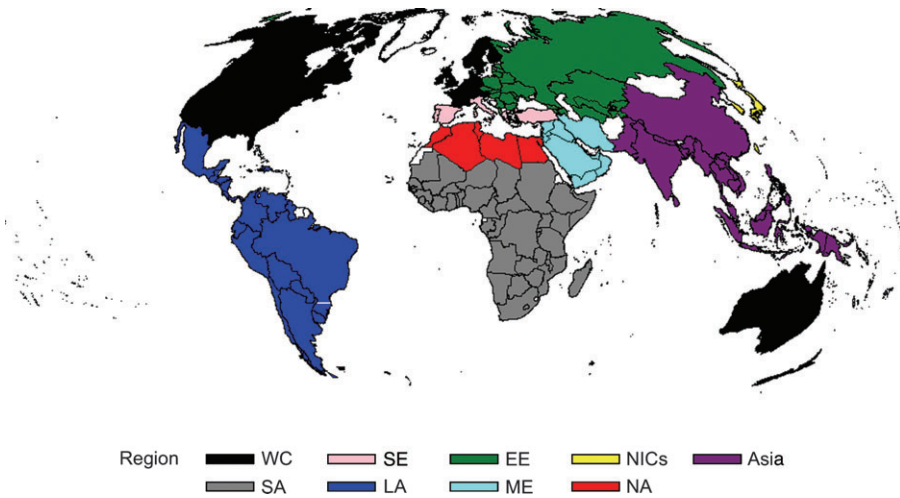
1. ACCOUNTING FOR GROWTH AND THE DATA

The data set that we use in this article has more depth and breadth than previously available data used for growth analysis. The analysis of growth requires many years of data. To be sure that relatively high-frequency phenomena such as business cycles are not affecting the outcome, a single growth observation should cover at least 10 years, and more likely 20 years.

The heavily used data set provided by Summers and Heston (1988, 1991, 2000) is a large cross section with limited time-series information. Summers and Heston's data set

1. A detailed data appendix that provides information on our data sources is available on request and online at the Web sites www.jerrydwyer.com and <http://people.clemson.edu/~rtamura>.

FIGURE 1



contains information on 152 countries but no information on any country prior to 1950. At 10-year intervals, Summers and Heston's data set has 487 observations. Their data set also has one important deficiency: It does not contain information on human capital. Information on human capital is available from Barro and Lee (1993), which contains information only since 1960. The merged data set using the information on human capital available in Barro and Lee contains 397 observations. At 20-year intervals, the number of observations falls to about 200 since 1960. Our data set includes quite a few more observations: 764 observations at 10-year intervals and 382 observations at 20-year intervals. Although our data set contains 145 countries instead of 152 countries, we observe per worker values of output, physical capital and human capital for each country for an average of 58 years. Our data extend backward beyond 1900 for 23 of the 145 countries.

Maddison (1995) provides an alternative long-coverage data set. A drawback to Maddison's data set is that it primarily contains information on per capita output. Though informative for some purposes, there are no corresponding data on aggregate inputs: physical and human capital. Maddison includes data on the physical capital stock for six countries and a function of years of schooling for three countries but no data on human capital. As a result, Maddison's

data themselves are not sufficient to analyze the relationship between output and inputs including human capital.

The countries in our data include 98% of the population of the world in 2000 (World Bank 2004). For each of these countries, we calculate output per worker, physical capital per worker, and human capital per worker. In this article, we summarize many of our findings with results by region in addition to reporting results for each country. Each of the 145 countries is included in one of nine regional groups. Figure 1 shows the countries included in our analysis and the groups in which they are included. Although these groupings are arbitrary to some extent, the basic criteria are geographic proximity and data availability for similar durations.²

Our primary source of data is B. R. Mitchell's three volumes, *International Historical Statistics* (1998a, 1998b, 1998c). Mitchell provides data on income, the labor force, population, the demographic breakdown of the population by age groups, investment rates, and school enrollments, all of which we use in our investigation. We update these data to more recent years and supplement them by data from Maddison (1995), Summers and Heston (2000), and the *World Development Report 2002* (World Bank 2001). These data

2. Our groupings are similar to those in Lucas (1998).

are used to calculate per capita output in 1985 international dollars, per worker output in 1985 international dollars, the stock of physical capital per worker in 1985 international dollars, and the average level of education and experience acquired by people employed.³ Though subject to measurement error, these data provide information on a broad set of countries over periods that cover about all the years possible with currently available data. As a check on the reliability of these data, we compare the overlapping years of the data set with other existing data sets.

For overlapping years, our data are highly correlated with existing data sets, which gives us more confidence than otherwise that the data for the nonoverlapping years are reasonable measures of the variables they are meant to represent. Our numbers on output per worker share original data sources with Summers and Heston's data; hence our finding a correlation that is essentially unity reassures us that there are no dramatic transcription differences but is not very informative. More informative is the correlation of 0.97 between the overlapping values of our real output per person (which equals real income per worker) and Maddison's values of real income per person, which suggests that differences between these estimates are relatively unimportant. The investment numbers underlying the physical capital stock also are from the same underlying sources as Summers and Heston's data, so this correlation again mainly assures us that transcription differences do not loom large. The correlation of overlapping estimates of our average education and Barro and Lee's estimates is 0.84 (25 and older) and 0.85 (15 and older).

Growth Accounting Framework

We use income per worker rather than the more usual measure of economic growth, income per person, as do recent contributions

by Mankiw et al. (1992) and Klenow and Rodriguez-Clare (1997a).⁴ We assume that the relationship between output and resources can be summarized by an aggregate production function which can be written

$$(1) \quad Y(t) = A(t)F(K(t), H(t)),$$

where $Y(t)$, $K(t)$, and $H(t)$ are output, physical capital, and human capital at t , and the parameter $A(t)$ represents the level of technology, TFP, at t . Writing the production function this way restricts changes in the production function to Hicks-neutral changes in TFP. If social marginal products equal private ones and there is perfect competition, equation (1) implies that

$$(2) \quad a = y - \alpha k - (1 - \alpha)h,$$

where α is capital's share of income and a lowercase letter denotes the growth rate of a variable per worker. While the factor shares, α and $1 - \alpha$ generally vary over time, we assume that such variation is relatively unimportant for our estimates.⁵ The growth rate of TFP, a , in equation (2) is a residual computed from the other variables which are observable. We use equation (2) to estimate the growth rate of TFP as well as the variation in its growth over time and across countries. Almost all estimates of the importance of TFP include some adjustment for expansion of the economy. Some attempt a further adjustment. For example, Hall and Jones (1999) present estimates using the capital-output ratio, a formulation suggested by Solow's model with endogenous physical capital growth and exogenous technological and human-capital growth. Hall and Jones's decomposition based on equation (2) is $y = [\alpha/(1 - \alpha)](k/y) + h + [1/(1 - \alpha)]a$. This decomposition attributes changes in output per worker to changes in the capital-output ratio, human capital per worker or to technology, implicitly supposing that these are the exogenous factors causing growth. It is

3. We convert each currency's real value into international dollars using Purchasing Power Parity exchange rates calculated by Summers and Heston in the overlapping years. An international dollar is a dollar with the same purchasing power in a given year over aggregate U.S. gross domestic product (GDP) as a U.S. dollar, but with a purchasing power over subaggregates and over detailed categories determined by average international prices rather than by U.S. relative prices (World Bank 2001).

4. Kuznets (1966, p. 1) defines economic growth as a sustained increase in output per person or per worker.

5. The framework does not assume that the aggregate production function is Cobb-Douglas, although the end result is the same as assuming that on the front end. We interpret the results as reflecting production that is not necessarily Cobb-Douglas but with TFP reflecting variation in inputs' income shares.

not self-evidently desirable or plausible to make such exogeneity assumptions.⁶ We think it is more likely that human and physical capital and technology all have endogenous and exogenous components, which means such transformed numbers can be less informative than numbers not so transformed. Table 3 provides the means used to calculate contributions in either decomposition. It is easy to mentally compute the traditional contributions from the means in Table 3, which appears later. Appendix Table A2 provides the estimated contributions of growth of the capital-output ratio, human capital growth and TFP growth to output growth.

The TFP growth rate in equation (2) need not represent only technological change and may not represent technological change at all. Measurement errors in output and physical and human capital can appear in TFP growth. For example, new physical capital with zero marginal product, such as a useless road, increases the measured growth of physical capital, but a marginal product of zero implies that output does not change; the resulting change in TFP is the negative of capital's factor share times the new physical capital. Changes in hours worked, such as the decline in the United States in the twentieth century, will show up as reductions in the growth of TFP, but Huberman's (2004) analysis indicates the deviations in changes across countries are too great to make any simple adjustment feasible. Barro (1999) shows that deviations of social and private marginal products can but need not result in terms included in TFP growth, as can increasing returns. In addition, changes in property rights and economic regime can result in apparent TFP changes, although such changes can be interpreted as changes in the

difference between the social and private marginal products. Furthermore, changes in technology can be reflected in the growth of physical and human capital. In short, there are many possible explanations of changes in TFP.

For our estimates in this article, we use a capital share α equal to 0.33. This is in the range of the careful cross-country estimates by Gollin (2002) of 0.25 to 0.35. Using a common capital share for all years for all countries may seem like a dramatic restriction. If we limited the analysis to countries for which we can reliably estimate income shares, however, the analysis would use a small fraction of the available data on output and input growth. The data requirements in many countries would be overwhelming because it would be important to separate sole proprietors' income, including farmers' income, into labor and capital components. It is not obvious that the errors introduced by such estimates of labor and capital income would be less than those introduced by using common income shares across countries.

Output per Worker

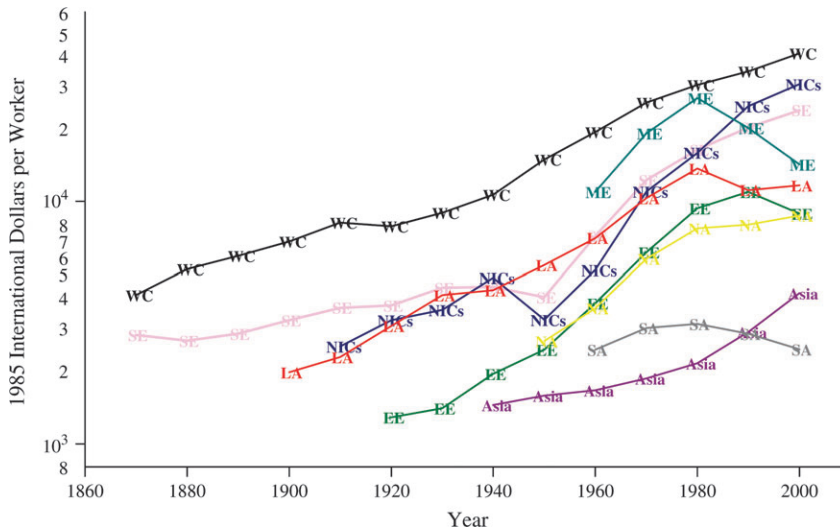
Mitchell (1998a, 1998b, 1998c) provides both nominal and real income per person through 1992. We use the overlapping years with the Summers and Heston data set to calculate real exchange rates between the local currencies and the Summers and Heston (1991) values in 1985 international dollars. We then apply this real exchange rate back through time. For 2000 incomes, we use the values in the *World Development Report 2002* (World Bank 2001) and convert these values to 1985 international dollars using the U.S. GDP deflator.

Our analysis focuses on the growth rates of output and inputs relative to the labor force. For simplicity, we often will refer to, for example, output per worker, by which we mean output per member of the labor force. Using output per worker instead of output per person simplifies the empirical analysis with no obvious loss in the informativeness of that analysis.

Figure 2 shows the behavior over time of the growth rates of output per worker for the nine regions. The slopes of the lines in Figure 2 are growth rates because the vertical

6. Suppose that the capital stock changes due to a gift from abroad or due to changes in financial intermediation, taxes on capital or the threat of expropriation. Will these changes permanently affect the growth of the capital-output ratio? The answer depends on tastes and technology. Klenow and Rodriguez-Clare (1997a) actually use the growth of physical capital relative to output and the growth of human capital relative to output, because they assume diminishing returns in these two produced factors of production. In our setup, $\alpha(k - y) + (1 - \alpha)(h - y) + a = 0$ because we have constant returns to scale in the two inputs. It seems to us better not to build such assumptions into growth accounting. We think that it is more informative to use the input means themselves. Bosworth and Collins (2003, pp. 131–33) reach a similar conclusion based on related arguments.

FIGURE 2
Output per Worker



Sources: See text and Data Appendix available online at www.jerrydwyer.com and <http://people.clemson.edu/~tamura>.

scale is proportional. To get reliable estimates of the growth rates, we have to undertake some involved computations. The value in the figure for output per worker in 2000 is the weighted average of the countries' output per worker in 2000. The weights are the 2000 share of the labor force in the region, which gives countries with larger labor forces more weight in the region. We then compute the weighted average growth rate for 1990 to 2000; the level of 1990 output per worker in the figure is the 1990 level of output per worker implied by this average growth rate. We then compute the weighted average growth rate for 1980 to 1990 for the countries with data in both 1980 and 1990; the level of 1980 output per worker in the figure is the 1980 level of output per worker implied by this average growth rate. We apply this procedure for a region as long as we have data on countries in the region that are at least 50% of the 2000 labor force. With this estimation procedure, the growth rate of output per worker for every period always is the growth rate of output per worker for the countries for which we have data over that time period. On the other hand, the level of output per worker for any years other than 2000 and 1990, when we have data for all countries, is not necessarily the actual level for the countries for which

we have data.⁷ Besides not having data available for the same periods for all countries, we do not always have data for exactly the same year for all countries. When data are not available for a particular year, we use output per worker in surrounding years to interpolate the data.⁸ Although this procedure would be problematic for some purposes such as a time-series analysis of the data, it has no effect on any of our conclusions. Perhaps the most obvious side effect for our purposes is interpolation's smoothing of the growth rates.

In Figure 2, the region called the Western Countries always has the highest output per

7. If the figure simply showed the weighted average level of output per worker in each year, the growth rates generally would not be the same as the growth rates computed for countries for which we have data at both the beginning and end of a period. If countries added to the data set have output per worker higher or lower than the average for the other countries, the growth rate over the period would be higher or lower because of the addition of the countries. To see whether the 2000 weights affect our conclusions, we repeated the computations with 1990 weights, 1980 weights and the average of 1980, 1990 and 2000 weights and the differences were small.

8. When necessary, we assume a constant growth rate between the surrounding years and interpolate to obtain data for the precise year used in the figure. We use the same procedure for all the series other than schooling, for which we assume constant arithmetic growth.

worker.⁹ Some regions narrow the gap with the Western Countries, whereas other regions fall behind, a result similar to that emphasized by Quah (1996), Pritchett (1997), Jones (1997) and Lucas (2000). From 1870 to 1970, we think we see a tendency toward convergence across regions: Once growth begins in a region, income per worker tends to catch up to the regions with higher levels of output per worker. Perhaps most striking to us is that it is only over the past 20 years (1980 to 2000) in which we have witnessed a divergence across regions as output per worker in Latin America, the Middle East, and South Africa fell. The decreases in the Middle East output per worker reflect decreases in output per worker in Iran, which has 39% of the labor force in the Middle East, and Iraq, which has 13% of the labor force. The turmoil associated with the downfall of communism is associated with falling measured real output per worker in Central and Eastern Europe from 1990 to 2000. More surprising, at least to us, is the 15% decrease in real output per worker in Latin America from 1980 to 2000 and 21% decrease in Sub-Saharan Africa from 1980 to 2000.¹⁰ In the modern history represented in Figure 2, there is nothing similar to these decreases other than for the decade including World War II. The recent decreases in output growth have been noted by Rodrik (1999), Carpena and Santos (2000), Easterly (2001), Evrensel (2002) and others. There is no settled explanation, although Figure 2 makes it clear that the recent period for these regions is atypical compared to other times and places.

Physical Capital per Worker

We use the perpetual inventory method to calculate the capital stock per worker. We have data on investment for almost all years. To convert Mitchell's nominal investment rates into purchasing power parity (PPP) investment rates, we regressed Summers and Heston PPP investment rates on nominal investment rates, real per capita GDP, and an interaction of nominal investment rates and per capita GDP. From these estimates, we con-

structed PPP investment rates for years that were not covered by Summers and Heston's data. We compute the capital stock at the end of each decade by assuming that the ratio of investment to income is equal to the average value for available years in that decade. The annual depreciation rate is 7%, and the growth rate of output per year is assumed to be constant between observations.

Figure 3 shows the evolution of capital stocks per worker for the nine regions. As in Figure 2, we weight each country in each region by the size of its labor force relative to the total labor force in the region. In 2000, the NICs have the highest stock of physical capital per worker due to more rapid growth of their capital stock in the 1990s.¹¹ Some other regions also have higher growth rates than the Western Countries. Decreases in the measured capital stock per worker are not the sources of the decreases in output per worker in the Middle East, Latin America, or Sub-Saharan Africa from 1980 to 2000. The growth rate of capital is negative in the Middle East from 1980 to 2000, but the measured level still is only 0.26% lower in 2000 than in 1980. Although the growth rate of capital in Sub-Saharan Africa is slower from 1980 to 2000 compared with 1960 to 1980, it still is positive; therefore, measured growth in capital cannot account for the negative growth rate of output per worker in central and southern Africa from 1980 to 2000. The growth rate of capital per worker in Latin America also is lower from 1980 to 2000 than for earlier years, but the growth rate is positive and does not reflect the time series pattern of output per worker, which falls from 1980 to 1990 and increases from 1990 to 2000.

Years of Schooling, Experience, and Human Capital per Worker

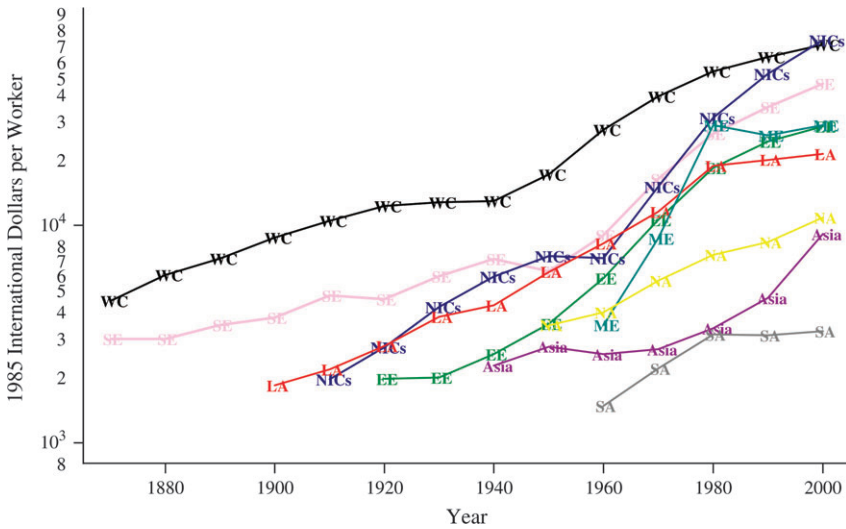
Our measure of human capital per worker in each country reflects both average education and average number of years employed. We compute education using formulas similar to those used by Barro and Lee (1993). The average number of years of schooling for an employed person is calculated from enrollments in primary and secondary schools and higher education in combination with the age distribution of the population. Enrollments

9. The region called the *Western Countries* includes Northern and Western European countries as well as the United States, Canada, Australia, and New Zealand.

10. There are no countries added to these regions after 1970, so compositional effects do not explain these differences.

11. The newly industrialized countries include Hong Kong, Japan, Singapore, South Korea, and Taiwan.

FIGURE 3
Capital per Worker



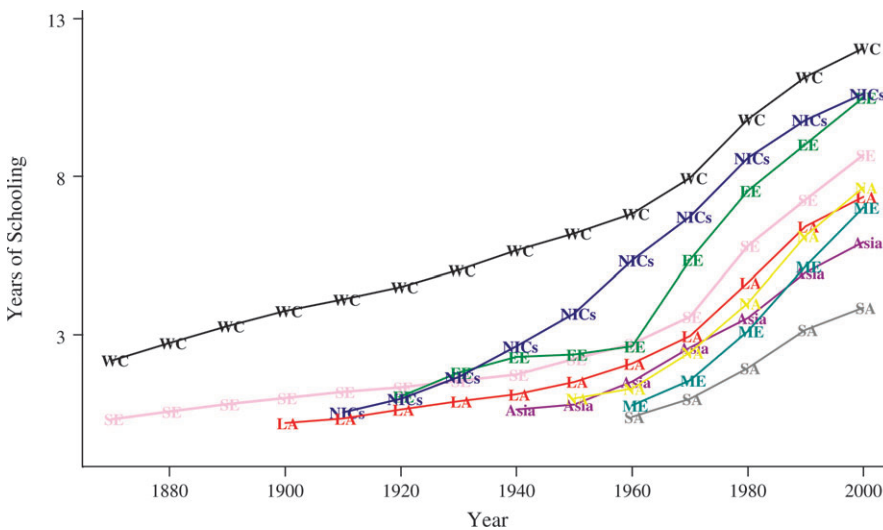
Sources: See text and Data Appendix available online at www.jerrydwyer.com and <http://people.clemson.edu/~tamura>.

are used to calculate the fraction of the population that has some primary schooling, some secondary schooling, and some college education. We use the age distribution of the population to estimate the age distribution of those employed because the data available to us do not include the age distribution of the

labor force. We also use the same level of education for men and women because we do not have enrollment data by gender.

Figure 4 shows the average years of schooling completed per worker for the nine regions. The Western Countries have a history of much higher education than the rest of the world.

FIGURE 4
Years of Schooling



Sources: See text and Data Appendix available online at www.jerrydwyer.com and <http://people.clemson.edu/~tamura>.

The Western Countries have an average education of 2.18 years in 1870, higher than the initial value for any other region, and this region has an average education of 12.06 years in 2000, the highest in the figure. Only since World War II has other regions' average education risen as high as the Western Countries' level of education in 1870.

Human capital per worker is computed from average education, Ed , and average experience, Ex . We do not have data on wages in the individual countries, which would allow us to compute contributions from increased education as suggested by Jorgenson and Griliches (1967). The transformation from educational attainment and experience to human capital instead is based on estimated parameters of earnings regressions. The evidence of substantial diminishing average returns to years of schooling indicates that it is important to distinguish between primary, secondary and higher education. Average years of schooling completed, Ed , is divided into years of primary schooling, P , years of intermediate schooling, I , and years of secondary and higher education, S .¹² We assume that primary school must be completed to attend intermediate school and that primary and intermediate school must be completed to attend secondary and higher school. We further assume that primary school attendance continues for up to four years, intermediate school attendance continues for up to four more years, and secondary and higher education continues for all later years. With these assumptions, knowing average attained education is sufficient to compute the average number of years of primary, intermediate, and secondary and higher education.¹³ We compute average experience, Ex , using Mitchell's demographic data as average age less average years of schooling and six years before attending school.¹⁴ With subscripts for country and year

suppressed for simplicity, human capital then can be computed from

$$(3) \quad H = H_0 \exp(\phi_P P + \phi_I I + \phi_S S + \lambda_1 Ex + \lambda_2 Ex^2),$$

where H is human capital, H_0 is the level of human capital with no schooling or experience; ϕ_P , ϕ_I , and ϕ_S are parameters on years of primary, intermediate, and secondary plus higher education; and λ_1 and λ_2 are parameters on years of work experience and experience squared.

We estimate human capital per worker using equation (3), relying on the estimates of wage regressions from different times and places for the estimated parameter values. Psacharopoulos (1994) summarizes the very large body of evidence on the relationship between wages and years of schooling across countries in the world. Following Hall and Jones (1999, p. 89), we use the following numbers as the point estimates: for the first four years of schooling, each additional year of schooling increases earnings by 13.4%; each additional year for the next four years of schooling increases the wage rate by 10.1%; and every year thereafter increases the wage rate by 6.8%.¹⁵ Klenow and Rodriguez-Clare (1997a) report estimates of the returns to education and experience from a cross-section of 48 countries with coefficients on experience and experience squared of 0.0495 and -0.0007. In sum, we use $\phi_P = 0.134$, $\phi_I = 0.101$, $\phi_S = 0.068$, $\lambda_1 = 0.0495$, and $\lambda_2 = -0.0007$ in equation (3).

Figure 5 shows human capital for the regions. The level of human capital itself does not have much content because there is a normalization associated with the level of human capital with zero schooling and labor force experience, H_0 . Hence, we set the level of human capital for the Western Countries to unity in 1870 and use the same normalizing constant for the other regions. The level of human capital is uniformly higher in the Western Countries, although the growth rates in other regions generally are higher. The percentage differences between the Western Countries and four other regions—the NICs, Central

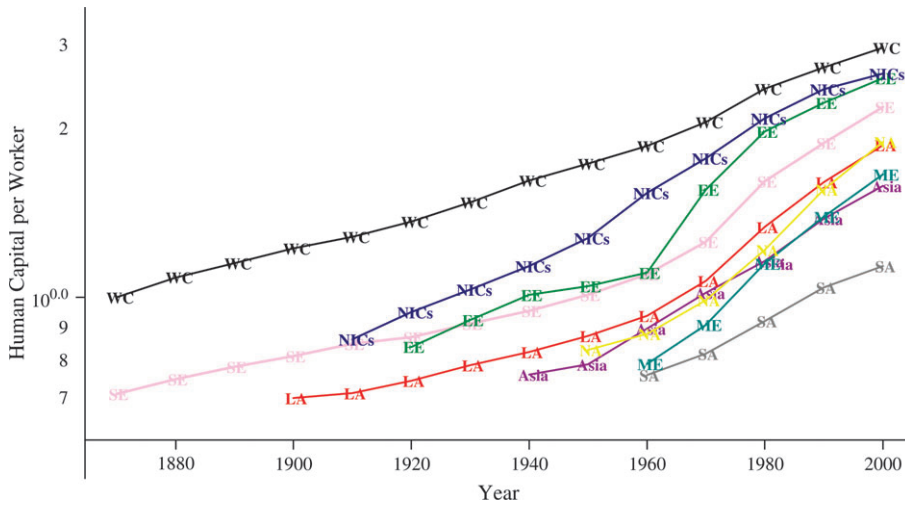
12. Higher education is not necessarily college education. Ed is the average number of years of school completed in a country and $Ed = P + I + S$.

13. If the average number of years of schooling is less than four years of schooling, then $P = Ed$, $I = 0$, and $S = 0$. If the average number of years of schooling is greater than four but less than eight, then $P = 4$, $I = Ed - 4$, and $S = 0$. If the average number of years of schooling is greater than eight, then $P = 4$, $I = 4$, and $S = Ed - 8$.

14. Knowing average years of schooling in the adult population, Y_s , and the average age of the population 6 to 64 not in school, Age , plus an assumption that school attendance begins at six years of age permits us to compute average years of work experience from $Y_w = Age - Y_s - 6$.

15. Psacharopoulos (1994) also presents estimates that vary by broad region of the world, which could be the basis of a more refined analysis.

FIGURE 5
Human Capital



Sources: See text and Data Appendix available online at www.jerrydwyer.com and <http://people.clemson.edu/~tamura>.

and Eastern Europe, Southern Europe, and Latin America—are less in 2000 than at any time earlier in the 1900s. After a dramatic growth rate from 1960 to 1970, the growth rate of human capital in the region with the least human capital—Sub-Saharan Africa—still is positive, but it actually is lower than for any other region. Hence, decreases in human capital are not behind the decreases in output in Latin America and Sub-Saharan Africa.

TFP Growth per Worker

Figure 6 shows the levels of TFP for the regions. TFP does not increase uniformly for any of the regions. Even for the Western Countries, the range of TFP growth rates over decades is from -1.24% per year from 1910 to 1920 to 1.98% per year from 1940 to 1950. Some other regions have more sustained decreases in TFP at times. The most sustained decreases in TFP are for Sub-Saharan Africa, a region that has negative growth of TFP at a rate of -1.82% per year for the 30 years from 1970 to 2000. Even before the decrease in real output from 1980 to 2000, Sub-Saharan Africa stands out in terms of having little growth and negative TFP growth. It is not necessary to suppose deteriorating technology to explain

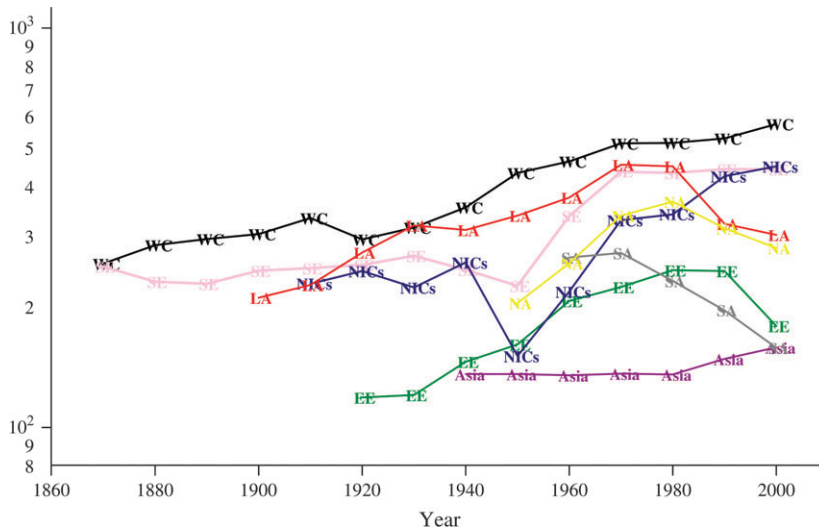
decreases in TFP. Many other factors, including decreases in competition in markets, increases in government regulation, and disruptions in private markets due to armed conflict, can account for these developments. That said, just as the decreases in real output in Sub-Saharan Africa are atypical, these large, sustained decreases in TFP are atypical. We conclude that the available data imply that it would be anachronistic to suppose that this divergence in real output and especially TFP across regions is anything other than a phenomena of the period since World War II.

II. GROWTH ACCOUNTING

How much of economic growth is associated with growth in aggregate input – physical and human capital weighted by factor shares—and how much with growth in TFP? Any average can be misleading. For all of our data, the weighted average growth rates of output per worker and TFP are 1.61% and 0.22% per year.¹⁶ Though this modest growth rate of

16. We use a weighted average across the countries, with weights equal to 2000 labor force and the number of years for which we have data rather than a simple average across countries. The reasons for this weighted average are explained in this section.

FIGURE 6
Total Factor Productivity



Sources: See text and Data Appendix available online at www.jerrydwyer.com and <http://people.clemson.edu/~tamura>.
Notes: The TFP values for the Middle East are: 1960, 872.56; 1970, 1026.63; 1980, 835.49; 1990, 581.86; 2000, 358.02. These values exceed the range of this graph and therefore have been left off for graphical purposes.

TFP seems a little surprising in light of Figure 6, we find it less surprising knowing that the growth rates of output per worker and TFP are 1.72% and 0.58% per year for the Western Countries—a nonnegligible growth rate of TFP relative to the growth of output per worker. For the United States, we estimate growth rates of output and TFP of 1.69% and 0.65% per year. For all countries, 14% of output growth is associated with TFP growth. For the Western Countries, one-third of output growth is associated with TFP growth. For the United States, 39% of output growth is associated with TFP growth.¹⁷

Comparison with Earlier Estimates

How different are our estimates than those made by others? There have been many breakdowns of economic growth into parts associated with aggregate input growth and parts associated with TFP growth. We compare our estimates with selected earlier estimates. Table 1 presents some influential estimates of output growth, input growth

and TFP growth by Abramovitz (1956), Solow (1957), Kendrick (1961), Denison (1985), Maddison (1995), and Abramovitz and David (2000). Different methods are used by these various authors. Still, the earlier estimates by Abramovitz, Solow, Kendrick, and Denison all indicate that there has been substantial TFP growth in the United States and that output growth bears little relationship with the growth of physical and human capital. Abramovitz's estimate is that growth of inputs accounts for 10% of output growth per person; Solow's is 12% of output growth per hour worked; Kendrick's is 20% of output growth per person; and Denison's is 32% of output per person employed. The numbers dramatically different than the others are those by Maddison—whose estimates are for total output, not for output per person or per worker—and Abramovitz and David (2000)—whose time period ends in 1989. Maddison's estimate is that growth of inputs accounts for 82% of total output growth in the United States from 1820 to 1992. In contrast to Abramovitz's (1956) finding that input growth accounts for 10% of output growth, Abramovitz and David (2000) find that input growth accounts for 57% of output growth per worker.

17. In the decomposition based on the capital-output ratio, TFP accounts for 21% of output growth for all countries and 50% for the Western Countries.

TABLE 1
Earlier Estimates of the Importance of Input and Productivity Growth for Output Growth

Investigator	Beginning Year	Ending Year	Income Measure	Labor Measure	Growth Adjustment	Growth Rate of Income	Growth Rate of Capital	Growth Rate of Labor	Growth Rate of TFP	Fraction of Growth Associated with TFP
Abramovitz	1869–78	1944–53	NNP	Hours	Per person	1.86	1.46	–0.08	1.68	0.90
Solow	1900	1948	Nonfarm private GNP	Hours	Hours	1.79	1.70	0	1.48	0.88
Kendrick	1889	1957	NNP	Weighted hours	Per person	2.0	1.2	0.1	1.6	0.80
Denison ^a	1929	1982	Potential NI	Weighted hours	Per person employed	1.48	1.34	0.16	1.01	0.68
Maddison	1820	1992	GDP	Hours worked	None	3.61	4.18	1.77	0.63	0.18
Abramovitz ^b and David	1890	1989	Private domestic product	Labor force	Labor force	2.49	2.23	0.92	1.08	0.43

Sources: Abramovitz (1956, table 1, p. 8); Solow (1957, pp. 315, 316); Kendrick (1961, table 8, p. 84 and table 29, p. 85); Denison (1985, table 3-4, p. 87; table 4-2, p. 93; and table 8-3, p. 113); Maddison (1995, tables K-1, K-2, pp. 253–55; table 2-5, pp. 41–42); and Abramovitz and David (2000, table 1.3, p. 14 and table 1.5, p. 20).

^aThe growth rates of capital and labor are growth rates on nonresidential businesses. Denison also includes land as a factor of production separate from capital.

^bAbramovitz and David present their estimates using hours, but we use their data on the labor force to make their estimates more comparable to ours.

Table 2 compares these estimates to ours. Our estimate for the United States is that TFP growth accounts for only 39% of output growth per worker, which is a far cry from the earlier estimates in Table 1. In Table 2, we decompose the differences between our estimate and these earlier ones into the pertinent factors. The first and last columns provide our estimate and these earlier estimates. The columns in between provide changes in TFP growth relative to output growth associated with the various factors. We start from our estimate. The factors that can account for differences in the fraction of output growth can be allocated reliably into differences in the time period, the growth rates of capital and human capital, and a residual category. This residual category reflects differences in growth adjustment, differences in income shares, possibly differences in the definition of income, and no doubt other differences. This residual category is not systematically large relative to the difference between our estimate and earlier estimates, which at the least means that these residual differences are not overwhelming.

Table 2 seems to split up in a natural way between the earlier estimates by Abramovitz and Solow, the later estimates by Kendrick and Denison and the recent estimate by Abramovitz and David. The differences between our estimates and those by Abramovitz and Solow are fairly evenly split between differences in time period, growth of physical capital and growth of human capital. Our adjustment for schooling and experience and Abramovitz's and Solow's lack of one is a major conceptual difference between our figures and theirs. The difference due to time period is interesting and large: TFP is 50% of growth using data up to either 1950 or 1953, and only 39% of growth if data through 2000 are included in the computations. If more recent estimates of investment are more accurate and our estimate of the implied capital stock is no worse, then capital growth is more important than they estimated. The differences between our estimates and those by Kendrick and Denison are less due to time period and the growth rate of human capital, although these differences remain.¹⁸ Differences

18. Kendrick includes growth of human capital through changes in relative earnings across industries. Denison (1985, p. 15) uses relative earnings to estimate the contribution of education to output and does not attempt to estimate the effect of experience.

TABLE 2
Our Estimates for Similar Time Periods of the Importance of Input and Productivity Growth for Output Growth

Investigator for Comparison	Our Estimate of Fraction of Growth Rate of Income Associated with TFP	Difference Due to					Fraction of Growth Associated with TFP
		Time Period	Growth Rate of Physical Capital	Growth Rate of Human Capital	Time Period and Growth of Aggregate Input	Other Factors	
Abramovitz	0.317	0.181	0.106	0.163	0.450	0.136	0.903
Solow	0.317	0.184	0.182	0.148	0.513	−0.003	0.827
Kendrick	0.317	0.063	0.273	0.114	0.450	0.033	0.800
Denison	0.317	0.057	0.388	0.072	0.517	−0.151	0.682
Maddison	0.150	−0.022	0.026	0.184	0.138	−0.113	0.175
Abramovitz and David	0.317	−0.052	0.068	0.124	0.140	−0.023	0.434

Note: Human capital estimates are the education and experience adjustments indicated in the text. Growth rates including TFP are per person in the labor force for the comparisons other than with Maddison, for which the growth rates are for the totals.

in the growth rate of physical capital loom relatively large. If we used Kendrick or Denison's estimate of the growth rate of physical capital with no allowance for conceptual differences associated with the differences in income measure, our estimates of the relative importance of TFP for their time periods would increase from 38% and 37%, respectively, to 65% and 76%. These estimates also would imply, though, that the ratio of physical capital to output decreases over time, a result that seems implausible and would be quite surprising (King and Levine 1994; Kuznets 1966). The difference between our estimates and Abramovitz and David's is primarily the contribution of human capital, which Abramovitz and David measure by estimating wages by age, sex, and education (Abramovitz and David 2000, p. 24).

Maddison estimates the importance of inputs for total output, and our estimates in Table 2 are estimates using output per worker. Hence, they are not directly comparable. Even so, there is less difference between our estimate and Maddison's than with the earlier estimates ending from 1948 to 1982.

Estimates of Average Growth and Total Factor Productivity Growth for Regions

Table 3 shows weighted and unweighted averages of the data for the regions and for the world.¹⁹ The weighted averages weight

the data for each country by the country's labor force in 2000 and the number of years for which we have data. The unweighted averages give the same weight to the smallest country, Guyana, with 254,000 workers, and the largest country, China, with 757 million workers! Averages weighted by labor force give more importance to the larger countries, which we think is helpful for interpreting the data. We also have big differences in the number of years for which we have data on countries. We have data on 16 of the 145 countries only since 1990 and for 93 of the countries only since 1950. We also have data for 23 countries for over 100 years. The unweighted averages give the same weight to each of the 23 countries for which we have data for 100 or more years and each of the 16 countries for which we have data for 10 years. Ten years is unlikely to be representative, especially if the country is new. Weighting by the number of years gives more weight proportionately to countries for which we have more data. These averages weighted by labor force and number of years can answer the question: What happened to the typical worker in a typical year for which we have data? Because unweighted averages also can be informative, we report them as well. Unweighted averages can answer the question: What happened to the typical country for the period for which we have data on each country?

Weighting is important for evaluating the overall average as well as for evaluating the averages for individual regions. The unweighted average growth rate of TFP in

19. The growth rates for individual countries are reported in Appendix Table A1.

TABLE 3
Average Growth of Output and Inputs by Region

Region	Growth Rate per Worker				TFP Growth Relative to Output Growth
	Output	Capital	Human Capital	TFP	
Weighted average					
All Countries	1.61	2.33	0.92	0.22	0.14
Western Countries	1.72	2.06	0.69	0.58	0.34
Southern Europe	1.75	2.18	0.86	0.46	0.26
Central and Eastern Europe	2.14	3.30	1.20	0.25	0.12
NICs	2.63	3.86	1.01	0.67	0.26
Asia	1.58	2.17	0.96	0.22	0.14
Middle East	0.98	5.64	1.70	-2.02	-2.07
North Africa	1.99	1.90	1.33	0.47	0.24
Sub-Saharan Africa	0.17	2.31	0.97	-1.25	-7.40
Latin America	1.59	2.27	0.86	0.26	0.17
Unweighted average					
All Countries	0.74	2.43	1.11	-0.81	-1.09
Western Countries	1.91	2.31	0.75	0.65	0.34
Southern Europe	2.57	2.99	0.97	0.93	0.36
Central and Eastern Europe	-0.84	2.02	0.98	-2.16	2.58
NICs	3.50	5.34	1.37	0.82	0.23
Asia	1.05	2.63	1.27	-0.67	-0.64
Middle East	0.09	3.81	1.60	-2.24	-24.88
Northern Africa	2.24	2.52	1.57	0.35	0.16
Sub-Saharan Africa	0.17	1.98	1.03	-1.18	-6.96
Latin America	1.23	2.21	1.19	-0.29	-0.24

Notes: When the new countries in Central and Eastern Europe in 1990 are deleted from the averages for that region, the average growth rates of output, physical capital, human capital, and TFP for that region are (1) weighted, 2.53, 3.43, 1.23, and 0.58% per year; and (2) unweighted, 3.61, 4.04, 0.96, and 1.63% per year.

Table 3 is an astounding -0.81% per year. This is quite inconsistent with our impression from the weighted averages in Figure 6, which has negative TFP growth rates but not predominately so. The weighted averages show that our overall impression of positive TFP growth is correct even if the weighted average is not exactly large: only 0.22% per year. Across all countries, our data support a statement that less than 15% of output growth per worker can be associated with TFP growth. For the Western Countries, though, weighted average TFP growth is 0.58% per year, which is about one-third of the growth of output per worker per year. For a few regions, the weighted averages show noticeable similarities in the share of growth in output per worker associated with growth in TFP: 34% for Western Countries; 26% for Southern Europe and the NICs, and 24% for North Africa. A plausible generalization for these

regions is that about one-fourth to one-third of output growth per worker is associated with TFP growth. Latin America falls somewhat below these regions with growth of TFP of 17% of output growth. The other regions with positive growth of TFP have similar TFP growth: growth in Central and Eastern Europe is 12%; and growth in Asia is 14%. The other two regions do not have positive TFP growth for the data available.²⁰

Overall, we conclude that TFP growth is a somewhat important part of average output growth per worker, but the lion's share of the growth in output per worker can be attributed to growth of aggregate input per worker even for the Western Countries, Southern Europe, the NICs, and North Africa. This finding is in line with Jones's (2002) finding for the United

20. Appendix Table A2 presents an alternative decomposition based on the capital-output ratio.

States that factor accumulation accounts for a large fraction of economic growth.²¹ For the rest of the world, TFP growth has been noticeably less important on net when it is even positive.

The weighting is important for the averages for some regions, perhaps most obviously Latin America and Central and Eastern Europe in Table 3. The unweighted average TFP growth rate for Latin America is negative and the weighted average is positive. The unweighted growth rates of output and TFP are negative for Central and Eastern Europe and the weighted average growth rates are positive. The individual data by country in Appendix Table A1 show why the weighting has these effects. In Latin America, the countries with substantial negative growth rates of TFP are small. Central and Eastern Europe has a large number of new countries in 1990, and the unweighted average is dramatically affected by these countries. The averages without these countries also are informative because 1990 to 2000 is a short period, and this is a period with substantial disruption for these countries. The remaining countries are Bulgaria, Czechoslovakia, East Germany, Hungary, Poland, Romania, Russia, and Yugoslavia. All weighted and unweighted average growth rates for these countries, most of which have data extending backward to the 1920s or earlier, are positive.²²

III. ESTIMATES OF VARIABILITY ACROSS COUNTRIES

Although TFP does not account for a large fraction of the average growth of output per worker, it may account for much of the variance across countries as argued by Klenow and Rodriguez-Clare (1997a) and Easterly and Levine (2001). They suggest that the variance of aggregate input growth explains almost none of the variance of output growth across countries; variance of TFP growth

explains virtually all of the variance of output growth across countries.

Possible Estimators of Relative Variances

What is an informative way to relate the variation of the growth rates? Let y be the growth rate of output per worker, x be the growth rate of aggregate input per worker, that is, $x = \alpha k + (1 - \alpha)h$, and a be the growth rate of TFP. By definition,

$$(4) \quad \text{Var}(y) = \text{Var}(x) + 2\text{Cov}(x, a) + \text{Var}(a)$$

which implies that

$$(5) \quad 1 = [\text{Var}(x)/\text{Var}(y)] + [\text{Var}(a)/\text{Var}(y)] + 2\rho_{x,a}[\text{SD}(x)\text{SD}(a)/\text{Var}(y)],$$

where $\rho_{x,a}$ is the correlation of the growth rates of x and a . If the correlation of TFP growth and aggregate input growth were zero, the first term would be the fraction of the variance of output growth due to variability of aggregate input growth, which also is the R^2 from a regression of output growth on aggregate input growth, and the second term would be the fraction of output growth due to TFP growth. In short, the least-squares decomposition would apply with TFP growth in the role of the residual. More generally, the least-squares decomposition does not apply because the correlation of TFP growth and aggregate input growth is not zero. As a result, it is impossible to uniquely estimate the fractions of output growth due to aggregate input growth and TFP growth absent some other assumption about the correlation of output growth due to aggregate input growth and TFP growth. One strategy would be to use the relative variances and ignore the covariance. This strategy would result in relative variances that do not add up to one if the correlation is non-zero; the relative variances can be greater than 1 if the correlation of aggregate input growth and TFP growth is negative. Klenow and Rodriguez-Clare (1997a) advocate a strategy of allocating one half of the correlation to each relative variance. This strategy creates relative variances that add up to 1, but the relative variances themselves can exceed one or be negative if the correlation is negative.

We take a different tack: We allow the data to inform us of plausible ways to allocate the covariance terms. This strategy yields two

21. Jones (2002) uses a model of growth of ideas to try to explain economic growth. He finds that, for the United States for 1950 to 1993, capital deepening, increased educational attainment and increased research and development (R&D) intensity account for 81% of the growth in this period. Our data for 1950 to 1990 using his number for R&D intensity indicate that 83% of growth is associated with input growth and R&D intensity.

22. Even though the region's average is affected substantially by deleting these new countries, the world average is little affected.

alternative estimates of the relative variances. These estimates alternatively attribute all of the correlation of aggregate input and TFP growth to either aggregate input or TFP growth. As a result, each of these estimates of the relative variances has a complement with which it adds up to 1. One way of explaining the underlying logic of these decompositions is statistical: The first decomposition assumes that all changes in output growth that are predictable by aggregate input growth are due to aggregate input growth; the second decomposition assumes that all changes in output growth that are predictable by TFP growth are due to TFP growth. Another way of stating it is in terms of unmeasured effects: The first decomposition assumes that the correlation of aggregate input growth and TFP growth reflects unmeasured effects of input growth on TFP; the second decomposition assumes that the correlation of aggregate input growth and TFP growth reflects unmeasured effects of TFP growth on inputs.

More formally, the first decomposition attributes to aggregate input growth all output growth predictable by aggregate input growth, which is consistent with a model of endogenous technological growth arising from capital accumulation such as those due to Romer (1986), Lucas (1988) and Tamura (1992, 2002, forthcoming). This decomposition can be written:

$$(6) \quad (\text{SD}(x) + \text{SD}(a)\rho_{x,a})^2 / \text{Var}(y) + (1 - \rho_{x,a}^2) \text{Var}(a) / \text{Var}(y) = 1.$$

The first term in equation (6) is the fraction of variation in output growth due to variation in aggregate input growth if all correlation of aggregate input growth and TFP growth reflects effects of growth in aggregate input. The second term is the fraction of variation in output growth not due to aggregate input growth; with this assumption about TFP growth, this fraction due to TFP growth is itself a fraction of $\text{Var}(x)/\text{Var}(y)$ that goes to 0 as $\rho_{x,a}$ goes to 1.²³

The second decomposition is the fraction of variation of output growth due to variation of

TFP growth if all correlation of aggregate input growth and TFP growth reflects effects of TFP growth. For example, the correlation might reflect differences in input growth rates induced by differences in TFP growth, which is consistent with the standard neoclassical growth model augmented to include exogenous technological progress. It is also consistent with endogenous technological change models, such as Romer (1990). Some insight into this decomposition can be gained from the representation:

$$(7) \quad (1 - \rho_{x,a}^2) \text{Var}(x) / \text{Var}(y) + (\text{SD}(a) + \text{SD}(x)\rho_{x,a})^2 / \text{Var}(y) = 1.$$

The second term in equation (7) is the fraction of variation in output growth due to variation in TFP growth if all correlation of aggregate input growth and TFP growth reflects growth in TFP. The first term is the fraction of variation in output growth not due to TFP growth; with this assumption about TFP growth, the fraction due to aggregate input growth is itself a fraction of $\text{Var}(x)/\text{Var}(y)$ that goes to zero as $\rho_{x,a}$ goes to one.²⁴

When the correlation of aggregate input growth and TFP growth is positive, these decompositions can be interpreted as alternative upper bounds on the importance of variation in aggregate input growth and TFP growth. The first decomposition attributes correlated variation in TFP growth to aggregate input growth; the second attributes correlated variation in aggregate input growth to TFP growth. A zero correlation poses no particular difficulties. In fact, our estimates would be the same as relative variances.

What if the correlation of aggregate input growth and TFP growth is negative though? The theories mentioned do not provide immediate support for plausible interpretations under these circumstances. One possible interpretation of a negative correlation is that there is a mistake in aggregate input growth that induces an opposite movement in TFP growth. For example, the construction of an unoccupied office building or an unused road would lead to a measured increase in capital

23. One way of seeing that the least squares decomposition holds for this representation is to note that the variance decomposition is $\text{Var}(y) = \beta_{y,x}^2 \text{Var}(x) + \text{Var}(e_{y,x})$ where $\beta_{y,x}$ is the regression coefficient from a regression of y on x and $e_{y,x}$ is the regression's residual.

24. The least squares decomposition holds for this representation also because the variance decomposition is $\text{Var}(y) = \text{Var}(y_{y,a}) + \beta_{y,a}^2 \text{Var}(a)$ where $\beta_{y,a}$ is the regression coefficient from a regression of y on a and $e_{y,a}$ is the regression's residual.

TABLE 4
The Relative Variability of Physical and Human Capital and Total Factor Productivity

Region	Number of Countries	Relative Variance of		Correlation of Aggregate Input and TFP	Variance Decomposition with All Correlation Associated with	
		Aggregate Input	TFP		Aggregate Input	TFP
All countries	145	0.14	0.73	0.20	0.30	0.86
Western Countries	16	0.13	0.45	0.88	0.89	0.97
Southern Europe	6	0.21	0.30	0.97	0.98	0.99
Central and Eastern Europe	24	0.05	0.73	0.59	0.53	0.97
NICs	5	0.53	0.17	0.49	0.87	0.60
Asia	16	0.61	0.84	-0.32	0.24	0.45
Middle East	10	0.85	1.00	-0.46	0.21	0.33
North Africa	5	1.07	0.30	-0.33	0.73	0.05
Sub-Saharan Africa	40	0.27	0.54	0.25	0.49	0.75
Latin America	23	0.18	0.74	0.11	0.27	0.82

and no change in output. In this case, measured TFP would necessarily fall. In fact, any aggregate input that is increased beyond its marginal product can lead to a fall in TFP because output does not rise as much as predicted. Such an outcome can be indicative of policies distorting markets so that resources are not allocated to their most efficient use. Another possible interpretation is that there is a common factor affecting aggregate input growth and TFP growth in opposite ways. Emigration of people with more human capital than the average person in the country is one possibility of such a factor: Emigration of people with more human capital would lower human capital used in production relative to our estimates and we would overestimate human capital used in domestic production. The precise interpretation of the decompositions would depend on details of the interpretation of factors that create the negative correlation.

Relative Variances for All Data

Table 4 presents estimates of the relative importance of aggregate input and TFP growth for the variance of output growth across countries.²⁵ The table is based on unweighted estimates of the variance of output growth across countries. We do not show weighted estimates of these numbers because

the purpose is to estimate the variance of output growth *across countries*, not across workers in the world.²⁶ We seldom use the phrase “per worker” because it gets repetitious, but all series in this and the next section are measured per worker, as have been all of the series presented thus far.

After the initial columns showing the region and number of countries, the first and second columns in Table 4 show the variances of aggregate input growth and TFP growth relative to output growth. These two measures will add up to unity only if the correlation of aggregate input growth and TFP growth is 0, which it generally is not. In fact, the sum of the variance of aggregate input and relative TFP exceeds one for Asia, the Middle East, and North Africa. For all countries, the relative variances of aggregate input growth and TFP growth add up to only 0.87, noticeably less than 1. This deviation from 1 is due to the positive correlation of aggregate input growth and TFP growth across all countries,

25. Appendix Table A3 presents the variance decompositions using capital-output ratios.

26. The unweighted statistics are informative about the variance across countries however large or small the countries may be, and the weighted statistics are more informative about the history confronted by the typical member of the labor force in our set of countries. Weighting by number in the labor force would be tantamount to attempting to estimate the personal distribution of income with these data. We see no useful purpose served by such an attempt. There might be a purpose to weighting the relative variances by years, but we see no reason to assume that variances are proportional to the number of years of data. We examine the relationship between the variances and the number of years in the next section.

shown in the third column of Table 4. There is substantial diversity in the correlation of aggregate input growth and TFP growth across regions, with a range from 0.97 for Southern Europe to -0.46 for the Middle East.

At first glance, the negative correlations of aggregate input growth and TFP growth for three regions seem odd. Examination of the individual countries suggests that country-specific explanations for these negative correlations may be important, at least in Asia and the Middle East. In Asia, Appendix Table A1 shows that Nepal and Vietnam have the second and fourth most negative TFP growth (-2.20% and -1.65% per year) and have the highest and fifth highest rates of aggregate input growth (3.13% and 2.07% per year). In the Middle East, Yemen and Saudi Arabia are the two countries with the most negative TFP growth (-7.84% and -2.28% per year), and these countries have the second and third highest growth rates of aggregate input— 6.61% and 3.28% per year.

The final two columns of Table 4 show our two estimates of the decomposition of output growth. These decompositions add up to 1 with the correct complement, which as a consequence is not shown. The decomposition in the next to last column of Table 4 assumes that the correlation of aggregate input growth and TFP growth reflects effects of input growth. The decomposition in the last column assumes that the correlation of aggregate input growth and TFP growth reflects effects of TFP growth.

For all the countries taken together, aggregate input growth is relatively unimportant compared to TFP growth. If all of the correlation of aggregate input growth and TFP growth reflects effects of aggregate input growth, as the next to last column of Table 4 supposes, then aggregate input growth accounts for 30% of the variance of output growth across these countries, with TFP growth accounting for the remaining 70%. Alternatively, if all correlation of aggregate input growth and TFP growth reflects effects of TFP growth, then TFP growth accounts for 86% of the variance of output growth across these countries and aggregate input growth accounts for 14%. Interpreting these two estimates as rough upper bounds on the importance of aggregate input growth and TFP growth, we conclude that these estimates suggest that TFP growth accounts for roughly 70–86% of the variation of output growth,

with aggregate input growth accounting for 14–30% of the variation of output growth.

Within regions the predominance of the importance of TFP growth appears to be confined to Central and Eastern Europe, Sub-Saharan Africa, and Latin America. For the Western Countries, aggregate input growth accounts for as much as 89% of the variance of output growth; in Southern Europe aggregate input growth accounts for as much as 98% of variance of output growth. The other regions in which the growth of aggregate input accounts for about 90% of output growth is the set of NICs, for which as much as 87% of the variance of output growth can be accounted for by aggregate input growth. For North Africa, aggregate input accounts for as much as 73% of output growth. For the regions with no more than 50% of the growth of output associated with growth in aggregate input—Central and Eastern Europe, Asia, the Middle East, Sub-Saharan Africa, and Latin America—none of them has an unweighted average growth rate of TFP in Table 3 that is positive.

These estimates do not suggest that the relative importance of TFP growth for the growth of output per worker across countries is solely due to the growth of technology. Instead, the relationship between negative TFP growth rates and the importance of TFP for the variance of output growth in a region suggests that other developments such as institutional changes, legal changes, and armed conflicts are the important ones for understanding why variability of TFP growth is a very important part of differences in growth experiences around the world.

The evidence concerning the relative importance of the variance of TFP growth and aggregate input growth in the Western Countries is not very informative. The upper bound estimate of each is on the order of 0.90. This means that the range of estimates of the importance of aggregate input and of TFP growth are from 10% to 90%. Someone with a strong prior about the importance of aggregate input or TFP growth of course would find little reason to revise that prior, essentially because there is a high correlation of aggregate input growth and TFP growth across these countries.²⁷ We start off with a diffuse prior, on the other hand, and find

27. Arguably that prior cannot be based on the data for the Western Countries or Southern Europe because we have the universe of available data for those regions.

TABLE 5
The Relative Importance of Physical and Human Capital and Total Factor Productivity for
Different Time Intervals

Time Period	Number of Countries	Time Interval	Standard Deviation of			Relative Variance of		Variance Decomposition with All Correlation Associated with	
			Output	Aggregate Input	TFP	Aggregate Input	TFP	Aggregate Input	TFP
1900–2000	25	1900–2000	0.525	0.281	0.341	0.285	0.421	0.654	0.766
		1920–2000	0.631	0.327	0.433	0.269	0.471	0.592	0.766
		1940–2000	0.697	0.416	0.436	0.357	0.392	0.652	0.683
		1960–2000	0.650	0.500	0.600	0.277	0.399	0.696	0.789
		1980–2000	1.126	0.527	0.963	0.219	0.731	0.272	0.845
1920–2000	34	1990–2000	1.784	0.666	1.414	0.139	0.628	0.429	0.878
		1920–2000	0.685	0.393	0.590	0.329	0.741	0.263	0.673
		1940–2000	0.743	0.434	0.617	0.340	0.688	0.313	0.660
		1960–2000	1.185	0.532	0.917	0.201	0.599	0.451	0.815
		1980–2000	1.928	0.593	1.791	0.095	0.863	0.141	0.906
1940–2000	47	1990–2000	3.044	0.767	2.797	0.063	0.844	0.190	0.939
		1940–2000	0.956	0.473	0.737	0.245	0.594	0.432	0.766
		1960–2000	1.416	0.554	1.085	0.153	0.587	0.523	0.876
		1980–2000	2.172	0.662	1.894	0.093	0.761	0.297	0.914
		1990–2000	3.297	0.924	2.953	0.079	0.802	0.243	0.926
1960–2000	118	1960–2000	1.899	0.830	1.447	0.191	0.581	0.490	0.832
		1980–2000	2.697	0.891	2.412	0.109	0.800	0.219	0.894
		1990–2000	4.266	1.340	3.897	0.099	0.834	0.177	0.903
1980–2000	128	1980–2000	2.694	0.918	2.399	0.116	0.793	0.225	0.887
		1990–2000	4.401	1.340	4.057	0.093	0.850	0.159	0.908
1990–2000	144	1990–2000	4.528	1.303	4.161	0.083	0.844	0.172	0.919

it plausible that the variance of the growth of aggregate input per worker and of TFP are equally important for explaining the variance of the growth of output per worker in the Western Countries. In some other regions and across all countries, however, variation in TFP growth is relatively more important.

Relative Variances for Common Periods

Our data cover quite different periods for the various countries, and this could affect our conclusion. In this section, we examine whether this possibility is correct. Because our data span a large number of countries for long periods, we also can examine another question: How long a period is necessary to draw reliable conclusions about the relative importance of aggregate input and TFP growth? It is obvious that a single year would be too short a period. Transitory developments and measurement error could overwhelm the long-term growth of the economy. Is 10 years enough? Forty years? Is 100 years necessary?

Table 5 presents estimates of the relative variance of output growth associated with aggregate input growth and TFP growth for common periods and shows the implications of the period length for the estimates. We start with the countries for which we have 100 years of data ending in 2000 and repeatedly chop 20 years off the beginning of the period until we hit 1980, at which point we chop off 10 years and compute the statistics for the last 10 years, 1990 to 2000.²⁸ The data for all countries end in 2000, and all periods in Table 5 end in 2000 in order to have roughly the same time period for each country. The number of countries included falls as the time period lengthens because many countries do not have data for as many as 100 years, although the

28. We did similar computations starting with various lengths of periods, for example, 100 years, 80 years, and so on, starting at the beginning of the data for each country. We found that effects of specific time periods explained numerous aspects of the statistics, at which point we shifted to the computations in the text.

computations for 100 years do include 25 countries. (We include India and Taiwan for which we have 99 and 95 years of data.) The first three columns show the time period, the number of countries, and the interval in the overall time period.

The fourth through sixth columns of Table 5 show the standard deviations of the growth of output, aggregate input, and TFP. All of these standard deviations increase as data over shorter periods are used to calculate the standard deviations, which is consistent with transitory developments being more important over shorter periods. Similar to columns 3 and 4 in Table 4, the next two columns of Table 5 show the simple estimates of the relative variances and the last two columns of Table 5 show our estimates of the importance of the variance of the growth of aggregate input and TFP.

TFP growth appears to be a substantial part of the variance of growth of output over periods of any length. TFP growth is as much as 77% of the variance of output growth for the 100 years from 1900 to 2000. This fraction is about the same for 100 years as it is for 40 or even 20 years for the same set of countries. It would be wrong, though, to conclude that the growth of aggregate input is unimportant.

For the 25 countries for which we have data for at least 100 years, growth of aggregate input also appears to be a substantial part of the variance of the growth of output: as much as 65% for the whole period. This fraction changes little, rising or falling a little, until the period is shortened to less than the last 40-year interval, 1960 to 2000, of the 100 years available for these countries.

The relative importance of the growth of aggregate input and TFP changes relatively little as the period is shortened and more countries are added, at least until the set of observations is broadened to include the 129 countries with data for at least the last 20 years. For the period 1980 to 2000, the growth of aggregate input appears to be a small part of the variance of the growth of output compared to the growth of TFP. This conclusion follows for the period 1990 to 2000 as well, with the exception of the countries with at least 100 years of data.

The estimates of the relative importance of the variance of aggregate input growth and TFP growth indicate that the decompositions generally are not sensitive to the time period or

their length. As would be expected, the variance of TFP growth, a residual in the computations, falls as the period length increases. The variance of output growth and aggregate input growth also fall, with the relative variances little affected until the period length is shortened to the 20 years from 1980 to 2000.

IV. CONCLUSION

Our new set of data covering 145 countries over a long time span provides evidence that little of the average growth of output per worker across the world is directly due to the growth of TFP: 14% for all of the countries. This conclusion, however, reflects substantial variance across countries—TFP accounts for about 34% of the average growth of output per worker in the Western Countries and 26% in Southern Europe and the NICs. Other regions have less, negligible, and even negative growth of TFP. These negative growth rates are consistent with the importance of institutional changes and conflicts. Our evidence indicates that, over long periods of time, the growth of output per worker is associated with accumulation of physical and human capital and technological change. At first glance, this conclusion might seem innocuous at best, but it is controversial in indicating that the growth of physical and human capital is important for growth.

Variation of the growth of aggregate input per worker and of TFP growth also are important in accounting for variation in the growth of output per worker. For all of our data, we conclude that variation in TFP growth is substantially more important than variation in aggregate input growth. There are interesting patterns by region though that are informative. We conclude that the variance of the growth of aggregate input and TFP are roughly equally important for Western Europe and Southern Europe. For the regions with negative average TFP growth rates, variation in the growth of TFP is substantially more important than variation in the growth of aggregate input per worker. This result is consistent with these negative growth rates being associated with institutional changes in some countries that have negative effects on output per worker in those countries and with armed conflicts involving some but not all countries.

At least with the data currently available, our evidence suggests that growth analysis

with less than a 40-year span may reach erroneous conclusions. We find that an analysis based on data for the last 20 years—1980 to 2000—would reach quite different conclusions than one based on the last 40 years—1960 to 2000.²⁹ A seemingly innocuous presumption that 20 years is long enough for analysis of growth would be wrong, at least for this particular period.

Our data span a long enough period that they can be used to address interesting, detailed questions. For example, we (Baier et al. 2004) have examined the effect of introducing stock exchanges on growth rates for 20 years after a stock exchange compared to 20 years before a stock exchange opens. We find that economic growth increases after a stock exchange opens in a country, primarily by increasing TFP growth. We are in the process of exploring whether other institutional changes and more general financial development and financial repression affect the growth of output, aggregate input, and TFP. Our results suggest that institutional developments, emphasized by North (1988), Grier and Tullock (1989), and Hall and Jones (1999), and possibly disruptions associated with armed conflict are important determinants of economic growth. Our data make it possible to examine the ability of such developments to explain why some countries grow and some countries do not, and also why even the countries that have economic growth on average sometimes grow and sometimes do not.

APPENDIX

For each country, Appendix Table A1 presents the average growth rate of output, physical and human capital per worker, and the average growth rate of TFP. The years spanned by our estimates also are indicated in the table.

Some recent growth accounting, for example by Klenow and Rodriguez-Clare (1997a), has used an alternative decomposition in which capital relative to output is used to measure the contribution of capital. With the same notation as in the article, this decomposition is

$$(A1) \quad y = \alpha(1 - \alpha)^{-1}(k - y) + h + (1 - \alpha)^{-1}a.$$

In this decomposition, the contribution of capital to TFP growth consists only of capital growth associated with changes in the capital-output ratio. This decomposition does not include capital growth that keeps pace with output growth as a contribution of capital growth to out-

put growth. This transformation increases the terms in human capital growth and TFP growth, which implies that the contribution of capital is less in equation (A1) than in the text. The logic underlying this change is illustrated by an example supposing that human capital growth increases. An exogenous increase in h increases output growth; if the capital-output ratio is to stay constant as it would in a steady state with a constant saving rate, the growth rate of capital therefore increases. Using equation (A1)'s decomposition avoids attributing some of this increase in output due to human capital growth to physical capital when the capital growth is simply an endogenous response to h and a . Suppose, on the other hand, that capital accumulation induces technological change and that such accumulation increases. This increases output growth and TFP growth. Equation (A1) underweights the contribution of capital growth in this example. Essentially, the issue is the relative endogeneity of k , h and a and whether it is desirable to try to sort out output growth due to or caused by these determinants. One interpretation of equation (A1) is that k/y , h , and a are being interpreted as exogenous determinants of growth. No doubt capital growth is partly an endogenous response to other things, but we think that the same can be said for all of the variables. Growth accounting is at best a summary of the data, not a causal or economic interpretation of the data, unless interpreted in the light of an explicit economic theory as in Jones (2002) for example. We interpret our empirical analysis in this article as summarizing a large amount of data that a solid theory would explain, similar to Parente and Prescott's (1993) summary of related data.

Appendix Table A2 summarizes the growth accounting by region using this transformation. The total contribution from TFP when the growth accounting is done this way is obtained by multiplying the contribution in Table 3 by $(1 - \alpha)^{-1} = (1 - 0.33)^{-1} = 1.493$. It is convenient to present average growth rates in Table 3, because only multiplication by 0.33 and 0.67 is necessary to get physical and human capital's contribution to output growth. In Appendix Table A2, it seems to us more convenient to see the contributions rather than the growth rates and that is what we present.

Presentation aside, this transformation raises the contribution of TFP to output growth to 21% for all countries when we weight by labor force years. The contribution increases to 50% for the Western Countries and to 39% and 38% for Southern Europe and the NICs. For the regions with negative TFP growth, the transformation makes TFP's contribution to growth more negative.

Appendix Table A3 presents the results of using this decomposition for the variance decomposition. This transformation induces high negative correlations between input growth and TFP growth, a point noted by Bosworth and Collins (2003, pp. 133–36). The correlation for the entire sample of countries is -0.68 . This is not entirely surprising. TFP is a residual based on output growth less physical and human capital growth. Growth of the capital-output ratio is the growth of capital less the growth of output. Changes in output growth that are not associated with input growth create a negative correlation between TFP growth and the growth of capital relative to output. Measurement error alone can produce this negative correlation, as can technological change that temporarily reduces the growth of capital relative to output, for example by increasing the growth of output and not affecting the growth of capital.

29. The past 40 years is the usable part of the Summers and Heston data when human capital is included in the analysis.

APPENDIX TABLE A1
Average Growth of Output and Inputs by Country

Country	Growth Rate per Worker				TFP	TFP Relative to Output
	First Year	Output	Capital	Human Capital		
<i>Western countries</i>						
Australia	1861	1.65	1.97	0.78	0.48	28.88
Austria	1880	2.04	2.44	0.79	0.71	34.58
Belgium	1846	1.95	2.38	0.70	0.69	35.59
Canada	1871	1.56	1.96	0.93	0.29	18.83
Denmark	1870	2.19	2.19	0.72	0.99	45.07
Finland	1850	1.53	1.90	0.83	0.34	22.34
France	1850	1.60	2.02	0.73	0.44	27.58
Germany	1880	2.55	3.00	0.78	1.04	40.64
Ireland	1926	3.87	4.04	0.92	1.91	49.52
Netherlands	1849	1.67	2.06	0.68	0.54	32.11
New Zealand	1911	1.97	2.38	0.82	0.64	32.48
Norway	1855	2.04	2.60	0.74	0.69	33.72
Sweden	1860	1.57	2.38	0.60	0.38	24.06
Switzerland	1888	1.63	2.13	0.69	0.46	28.49
United Kingdom	1831	1.08	1.59	0.71	0.08	7.03
United States	1870	1.69	1.89	0.61	0.65	38.75
<i>Southern Europe</i>						
Cyprus	1950	5.69	6.16	1.50	2.65	46.58
Greece	1910	2.77	3.39	1.02	0.97	35.04
Italy	1861	1.72	2.41	0.81	0.38	22.32
Portugal	1849	1.89	2.21	0.64	0.73	38.59
Spain	1857	1.33	1.67	0.72	0.29	22.05
Turkey	1935	2.03	2.08	1.14	0.57	28.29
<i>Central and Eastern Europe</i>						
Albania	1990	2.42	−2.15	0.43	2.84	117.38
Armenia	1990	−9.03	−1.28	0.62	−9.03	99.95
Azerbaijan	1990	−5.57	1.94	1.12	−6.96	124.96
Belarus	1990	−0.59	3.36	0.65	−2.12	363.05
Bulgaria	1934	2.27	2.86	0.83	0.77	33.91
Czechoslovakia	1921	3.62	4.56	0.85	1.55	42.79
East Germany	1962	7.03	10.65	1.07	2.80	39.84
Estonia	1990	2.24	4.37	1.02	0.11	4.77
Georgia	1990	−8.64	−3.44	0.63	−7.93	91.73
Hungary	1890	2.98	3.34	0.69	1.41	47.44
Kazakhstan	1990	1.25	2.84	0.97	−0.33	−26.56
Kyrgystan	1990	−1.61	−0.41	2.61	−3.22	200.09
Latvia	1990	−2.34	−1.92	0.51	−2.05	87.46
Lithuania	1990	0.68	2.52	0.83	−0.70	−103.60
Moldova	1990	−9.13	1.86	0.68	−10.19	111.70
Poland	1931	3.06	2.54	1.22	1.40	45.87
Romania	1930	4.55	4.54	1.22	2.24	49.11
Russia	1917	1.91	2.98	1.33	0.04	2.09
Slovak Republic	1990	7.10	6.53	0.76	4.43	62.43
Tajikstan	1990	−9.27	−2.98	1.55	−9.33	100.62
Turkmenistan	1990	−5.03	0.03	1.08	−5.77	114.56
Ukraine	1990	−5.06	2.04	0.41	−6.02	118.79
Uzbekistan	1990	−4.09	−0.92	1.19	−4.59	112.12
Yugoslavia	1920	1.16	4.57	1.22	−1.16	−99.92
<i>Newly Industrialized Countries</i>						
Hong Kong	1960	4.86	6.74	1.72	1.48	30.46
Japan	1890	2.53	3.43	0.91	0.79	31.17
Singapore	1963	4.53	7.27	1.89	0.86	19.09

continued

APPENDIX TABLE A1
Continued

Country	Growth Rate per Worker				TFP	TFP Relative to Output
	First Year	Output	Capital	Human Capital		
South Korea	1910	2.59	4.74	1.27	0.18	6.94
Taiwan	1905	2.98	4.54	1.07	0.77	25.69
<i>Asia</i>						
Bangladesh	1970	-1.09	-0.57	0.82	-1.45	133.18
Cambodia	1980	0.12	3.32	1.17	-1.76	-1469.65
China	1933	1.95	2.57	1.04	0.40	20.67
Fiji	1960	1.02	1.45	1.92	-0.75	-73.73
India	1901	1.30	1.57	0.76	0.27	21.05
Indonesia	1951	1.75	4.44	1.39	-0.65	-37.06
Laos	1980	-0.30	2.91	1.59	-2.33	772.72
Malaysia	1960	2.70	5.60	1.40	-0.09	-3.47
Myanmar	1941	0.15	-1.53	0.81	0.12	76.36
Nepal	1960	0.93	6.51	1.46	-2.20	-236.43
Pakistan	1951	1.08	2.12	0.85	-0.19	-17.57
Papua New Guinea	1960	0.75	2.37	0.88	-0.62	-83.09
Philippines	1939	1.95	1.92	1.54	0.28	14.44
Sri Lanka	1946	1.64	3.73	1.49	-0.60	-36.37
Thailand	1937	2.49	3.69	1.15	0.50	20.21
Vietnam	1980	0.43	2.00	2.11	-1.65	-386.92
<i>Middle East</i>						
Iran	1956	1.46	5.03	1.73	-1.37	-93.74
Iraq	1950	0.78	5.06	1.67	-2.01	-259.19
Israel	1948	3.17	4.66	1.69	0.51	16.01
Jordan	1960	1.02	4.59	1.12	-1.24	-121.39
Kuwait	1980	-1.11	-4.53	0.97	-0.26	23.56
Oman	1970	0.76	3.71	2.07	-1.85	-242.18
Saudi Arabia	1960	1.00	6.98	1.46	-2.28	-226.80
Syria	1953	0.64	3.02	1.91	-1.64	-256.99
United Arab Emirates	1980	-5.60	-6.64	1.45	-4.37	78.17
Yemen	1970	-1.24	16.20	1.88	-7.84	633.45
<i>Northern Africa</i>						
Algeria	1948	2.65	2.70	1.62	0.67	25.46
Egypt	1917	1.96	1.67	1.24	0.58	29.44
Libya	1960	2.91	4.60	2.09	-0.00	-0.11
Morocco	1951	1.35	1.69	1.22	-0.02	-1.83
Tunisia	1956	2.32	1.93	1.71	0.54	23.11
<i>Sub-Saharan Africa</i>						
Angola	1960	-1.99	-0.15	1.06	-2.65	133.25
Benin	1960	-1.35	1.02	1.02	-2.37	175.42
Botswana	1960	4.76	7.86	1.76	0.99	20.81
Burkina Faso	1960	1.57	3.99	0.48	-0.07	-4.66
Burundi	1960	-1.23	1.80	0.65	-2.26	183.69
Cameroon	1960	-0.50	2.65	1.29	-2.24	444.54
Central African Republic	1960	0.73	-0.28	0.94	0.20	26.69
Chad	1960	-0.81	2.45	0.69	-2.08	256.98
Congo	1960	-2.11	-0.10	1.95	-3.39	160.25
Ethiopia	1950	0.62	3.49	0.45	-0.83	-134.47
Gabon	1960	3.41	3.38	1.41	1.35	39.55
Gambia, The	1960	0.97	6.14	0.98	-1.72	-176.94
Ghana	1960	0.47	0.56	1.47	-0.70	-148.97
Guinea	1960	1.10	0.71	0.56	0.49	44.44
Guinea-Bissau	1960	-0.92	-0.32	0.72	-1.29	140.94
Ivory Coast	1960	0.37	0.91	1.07	-0.65	-177.95

continued

APPENDIX TABLE A1
Continued

Country	Growth Rate per Worker				TFP	TFP Relative to Output
	First Year	Output	Capital	Human Capital		
Kenya	1962	0.54	1.60	1.54	-1.01	-186.45
Lesotho	1960	5.17	12.71	1.42	0.03	0.55
Liberia	1960	-1.02	-0.01	1.09	-1.75	171.41
Madagascar	1960	-1.64	4.00	0.46	-3.27	198.90
Malawi	1960	0.26	1.33	0.67	-0.62	-236.12
Mali	1960	-0.78	2.10	0.45	-1.77	227.45
Mauritania	1960	0.43	0.43	0.85	-0.27	-63.00
Mauritius	1960	1.25	1.48	1.66	-0.35	-27.64
Mozambique	1960	-3.40	-0.29	0.69	-3.76	110.67
Namibia	1960	1.62	1.07	1.64	0.17	10.55
Niger	1960	0.11	0.46	0.39	-0.31	-287.94
Nigeria	1952	0.09	4.35	1.12	-2.09	-2244.35
Rwanda	1960	-1.29	-1.06	0.97	-1.59	123.38
Senegal	1970	0.32	1.66	0.98	-0.89	-279.04
Sierra Leone	1961	-2.30	5.19	1.08	-4.74	205.95
Somalia	1960	-1.41	1.01	0.42	-2.03	143.90
South Africa	1946	2.61	2.56	1.48	0.77	29.67
Sudan	1970	0.49	0.88	0.99	-0.47	-95.13
Tanzania	1960	0.29	3.15	0.77	-1.27	-443.86
Togo	1960	3.23	4.36	1.50	0.79	24.35
Uganda	1959	-0.53	0.62	0.81	-1.28	240.40
Zaire	1950	0.90	2.12	0.95	-0.43	-48.25
Zambia	1950	-3.50	-2.61	1.04	-3.33	95.29
Zimbabwe	1950	0.21	-1.87	1.68	-0.30	-144.49
<i>Latin America</i>						
Argentina	1895	1.60	2.17	1.04	0.19	11.71
Bolivia	1950	0.36	0.99	1.44	-0.93	-262.46
Brazil	1872	1.67	2.18	0.67	0.50	29.94
Chile	1895	1.53	2.21	0.98	0.15	9.61
Colombia	1917	1.25	1.97	0.97	-0.05	-4.06
Costa Rica	1951	2.49	3.14	1.47	0.47	18.76
Dominican Republic	1950	2.58	3.71	1.65	0.25	9.69
Ecuador	1950	1.03	3.35	1.73	-1.24	-120.66
El Salvador	1950	1.42	2.16	1.28	-0.15	-10.57
Guatemala	1950	1.34	2.83	1.04	-0.29	-21.72
Guyana	1946	0.92	2.17	0.82	-0.34	-37.18
Haiti	1950	1.39	2.94	0.93	-0.20	-14.36
Honduras	1930	0.24	1.37	1.00	-0.89	-375.33
Jamaica	1953	1.12	3.48	1.37	-0.95	-85.00
Mexico	1895	1.90	2.61	0.91	0.43	22.78
Nicaragua	1950	-0.12	0.51	1.34	-1.19	971.23
Panama	1950	1.68	2.94	1.50	-0.29	-17.32
Paraguay	1939	0.95	1.39	1.07	-0.23	-24.54
Peru	1908	1.42	2.28	1.03	-0.03	-1.78
Puerto Rico	1960	2.81	1.69	1.40	1.32	46.90
Trinidad	1960	-1.05	0.77	1.28	-2.16	205.93
Uruguay	1939	1.39	1.67	1.22	0.02	1.49
Venezuela	1936	0.47	2.25	1.26	-1.12	-237.90

APPENDIX TABLE A2
Contribution of Growth of Capital-Output Ratio, Human Capital and TFP to Average Output Growth

Region	Output Growth	Contribution of the Growth of Inputs		Contribution of the Growth of TFP	TFP Growth Relative Output Growth
		Capital-Output Ratio	Human Capital	TFP	
Weighted average					
All countries	1.61	0.36	0.92	0.33	0.21
Western Countries	1.72	0.17	0.69	0.86	0.50
Southern Europe	1.75	0.21	0.86	0.69	0.39
Central and Eastern Europe	2.14	0.57	1.20	0.37	0.17
NICs	2.63	0.61	1.01	1.01	0.38
Asia	1.58	0.29	0.96	0.33	0.21
Middle East	0.98	2.30	1.70	-3.02	-3.09
North Africa	1.99	-0.04	1.33	0.70	0.35
Sub-Saharan Africa	0.17	1.05	0.97	-1.86	-11.03
Latin America	1.59	0.34	0.86	0.39	0.25
Unweighted average					
All countries	0.74	0.83	1.11	-1.20	-1.62
Western Countries	1.91	0.20	0.75	0.96	0.50
Southern Europe	2.57	0.20	0.97	1.39	0.54
Central and Eastern Europe	-0.84	1.41	0.98	-3.22	3.85
NICs	3.50	0.91	1.37	1.22	0.38
Asia	1.05	0.78	1.27	-1.00	-0.95
Middle East	0.09	1.83	1.60	-3.34	-37.14
North Africa	2.24	0.14	1.57	0.52	0.23
Sub-Saharan Africa	0.17	0.89	1.03	-1.75	-10.38
Latin America	1.23	0.48	1.19	-0.44	-0.35

APPENDIX TABLE A3
The Relative Variability of the Capital-Output Ratio, Human Capital and Total Factor Productivity

Region	Number of Countries	Relative Variance of		Correlation of Capital-Output Ratio and Human Capital with TFP	Capital-Output Ratio and Human Capital	
		Capital-Output Ratio and Human Capital	TFP		Capital	TFP
All countries	145	0.26	1.62	-0.68	0.13	0.87
Western Countries	16	0.03	1.01	-0.11	0.00	0.97
Southern Europe	6	0.04	0.67	0.85	0.82	0.99
Central and Eastern Europe	24	0.12	1.62	-0.85	0.56	0.97
NICs	5	0.43	0.38	0.23	0.64	0.60
Asia	16	1.03	1.88	-0.69	0.01	0.45
Middle East	10	1.52	2.22	-0.75	0.01	0.33
North Africa	5	1.32	0.68	-0.53	0.51	0.05
Sub-Saharan Africa	40	0.31	1.21	-0.42	0.01	0.75
Latin America	23	0.32	1.65	-0.67	0.09	0.82

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